2021 Antelope fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Antelope (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 1 August 2021 at 17:00 UTC to 10 September 2021 at 02:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 942 hourly perimeters, a final area of 599.09 km^2 and an official size of 145,632 ac. The configuration uses 1 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

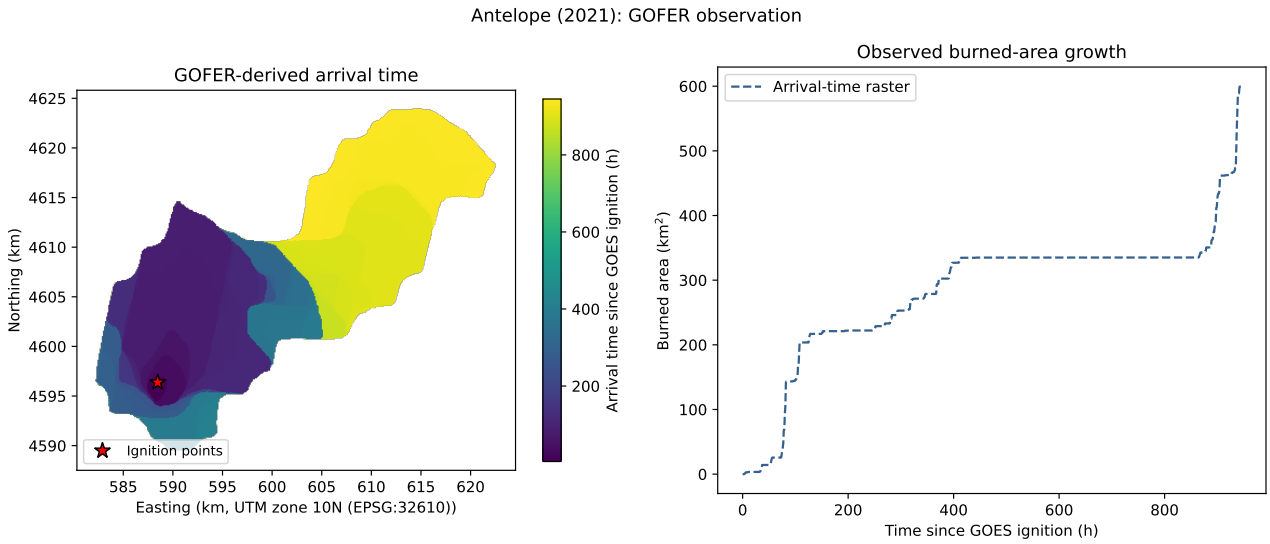


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,067 by 1,867 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 3,473 km^2. Elevation ranges from 1,175 to 2,618 m; median slope is 3°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 35%, and canopy base height has median 0.5 m.

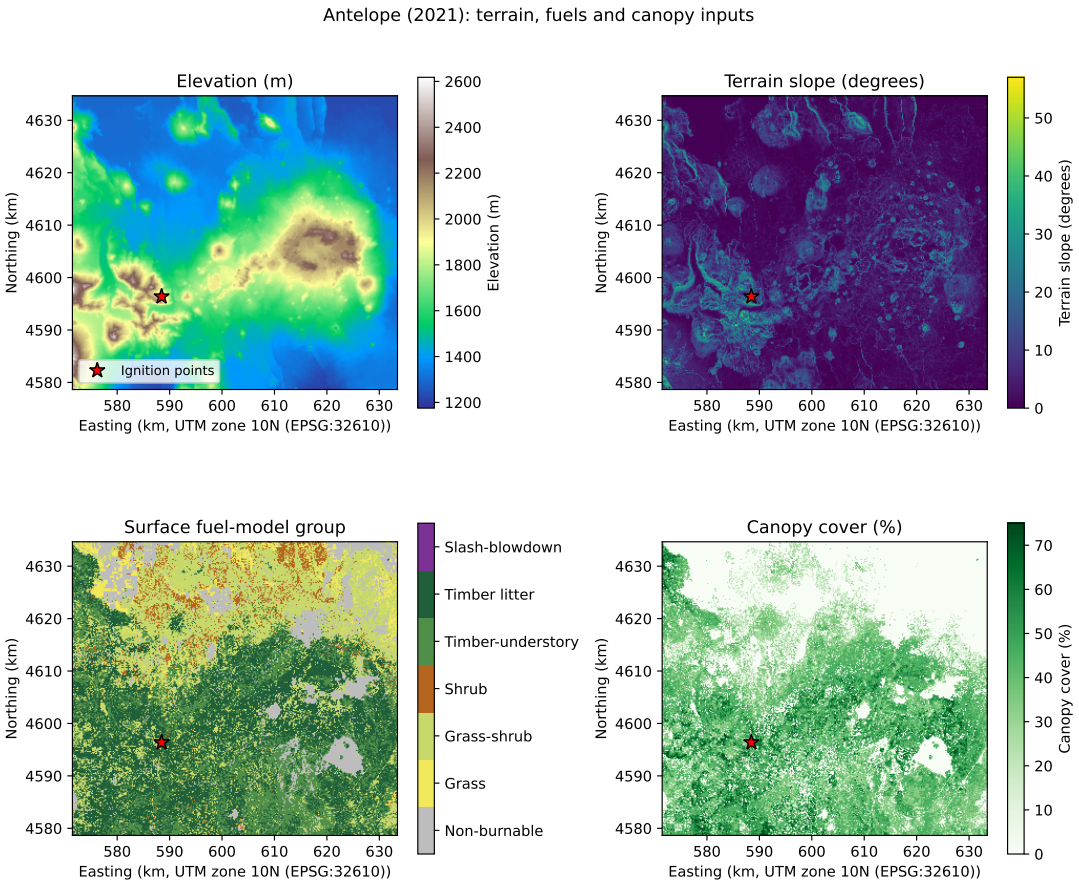


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

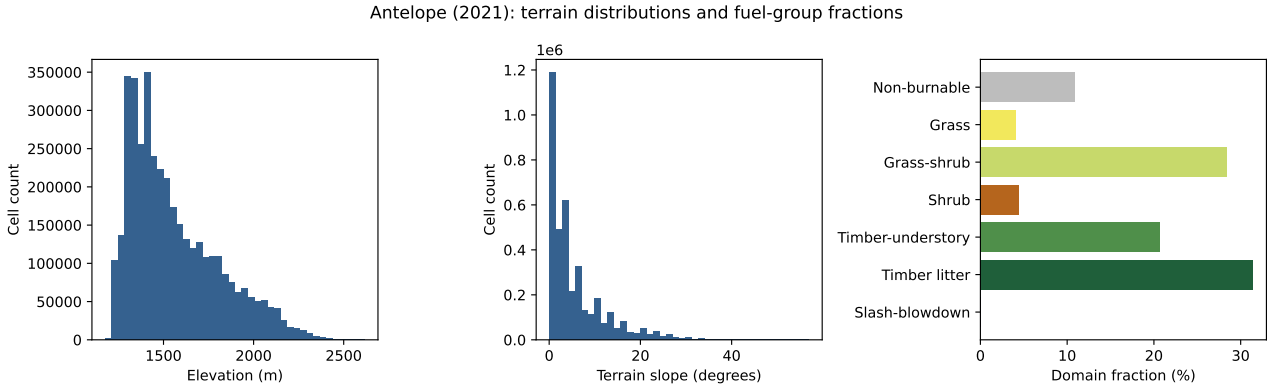


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 943 hourly fields. Median wind speed is 4.49 mph and its 95th percentile is 10.6 mph. Direction is summarized circularly: the mean direction is 293° and mean resultant length is 0.228. Median 1-h dead-fuel moisture is 5.92%. Spatial and temporal variability remain important when these domain summaries are interpreted.

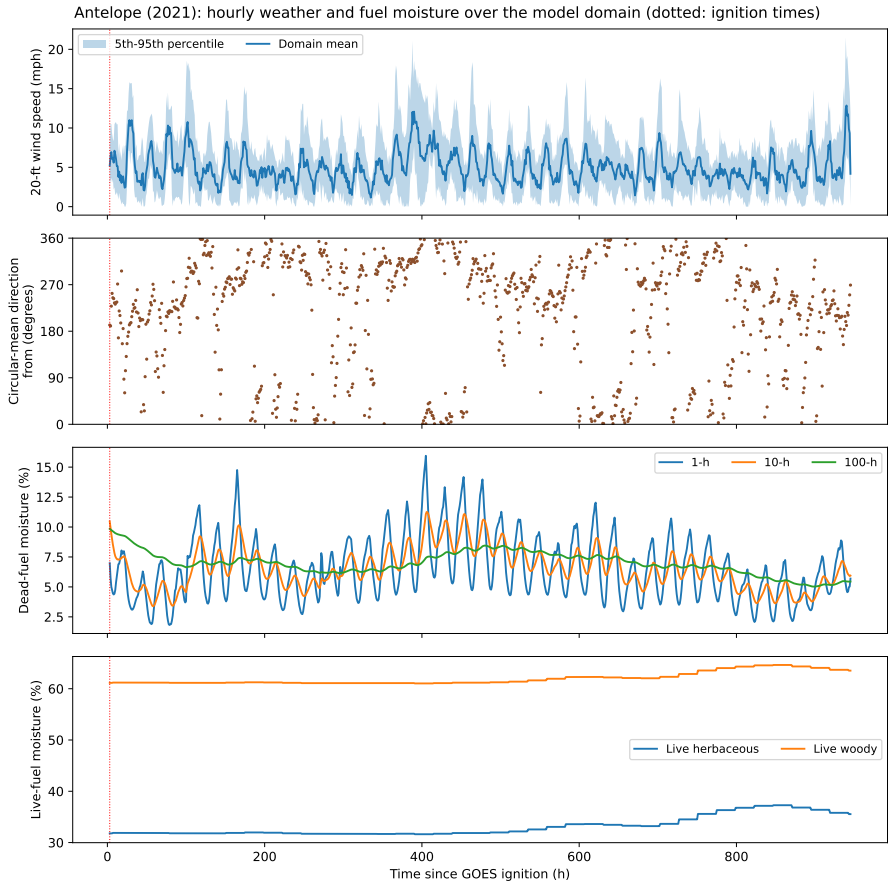


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

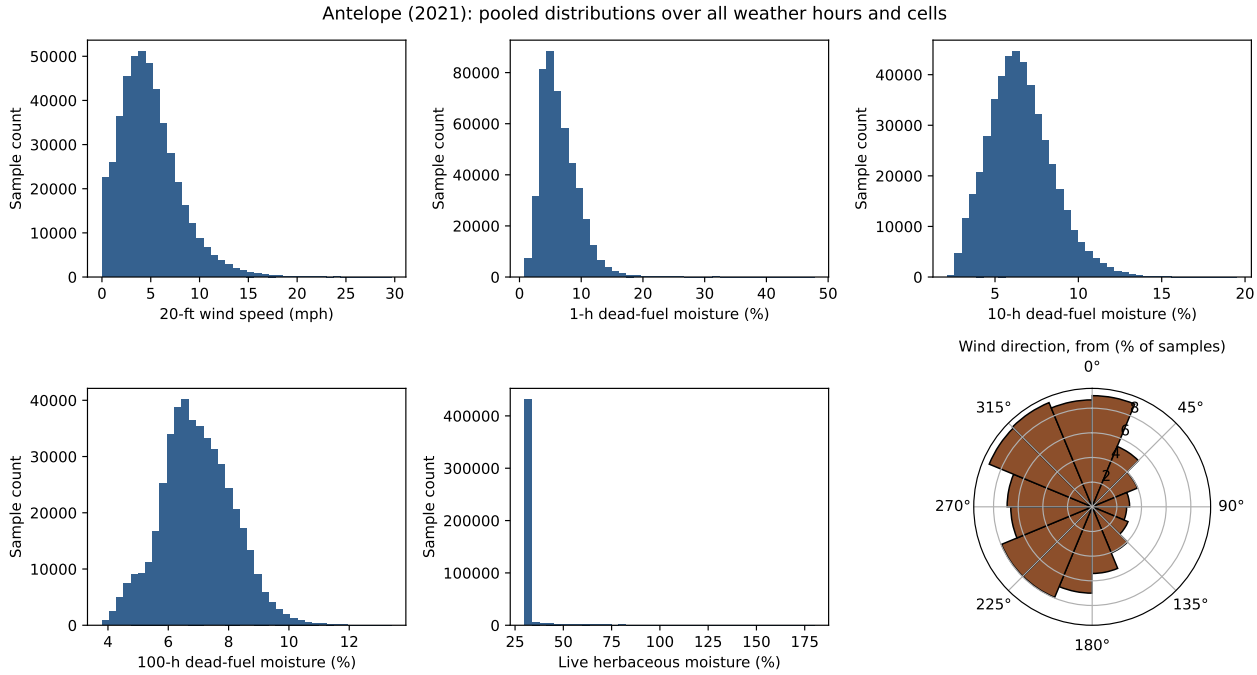


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 942 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 943 hourly weather fields and 1 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 3 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

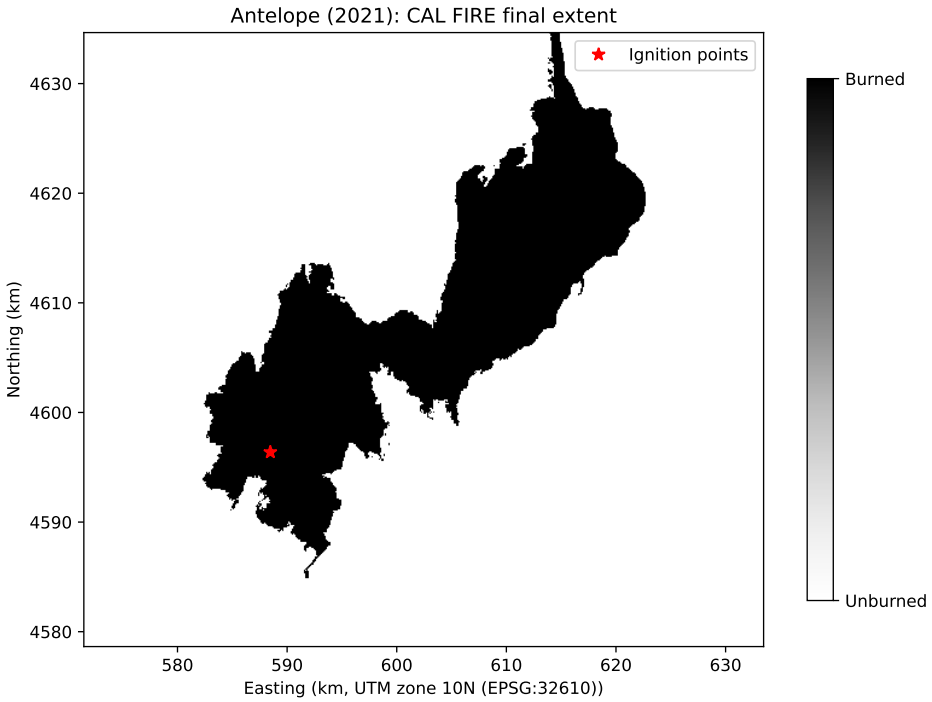


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 587 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available.

Current case configuration

Configuration description not documented in the available case materials.

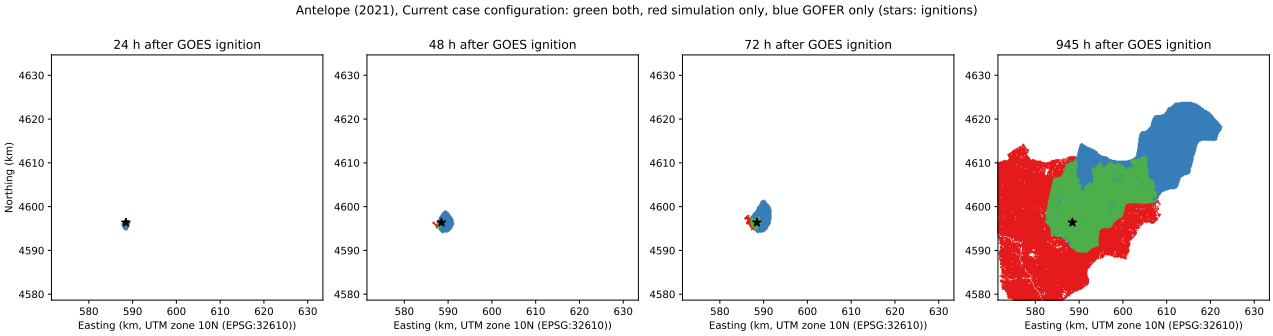


Figure 7: Current case configuration: simulated and GOFER-derived footprints at the fixed comparison times (hours since the GOES ignition hour) on the 30-m model grid: cells burned in both (green), only in the simulation (red), and only in the GOFER observation (blue). Stars mark the configured ignition points.

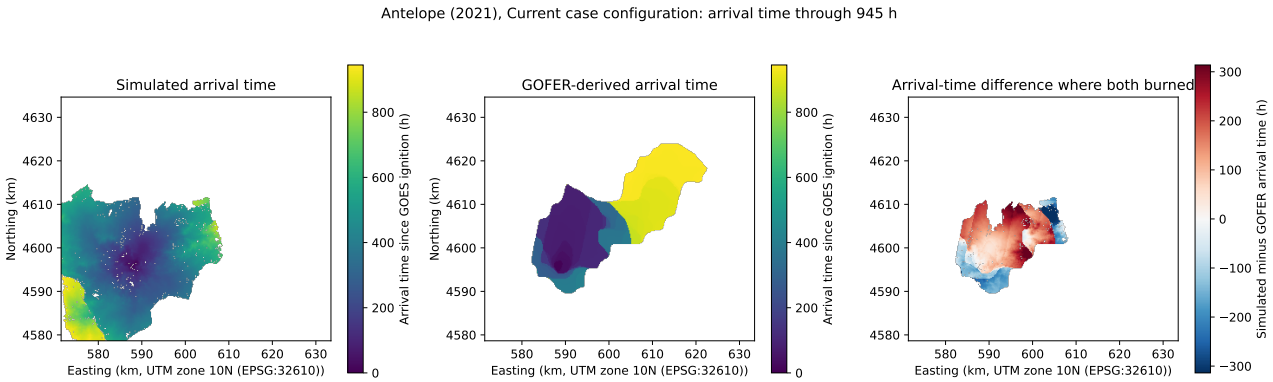


Figure 8: Current case configuration: simulated arrival time, GOFER-derived arrival time and their difference where both burned through the final comparison time, all in hours since the GOES ignition hour. The difference color scale is symmetric and clipped at the 95th percentile of its magnitude.

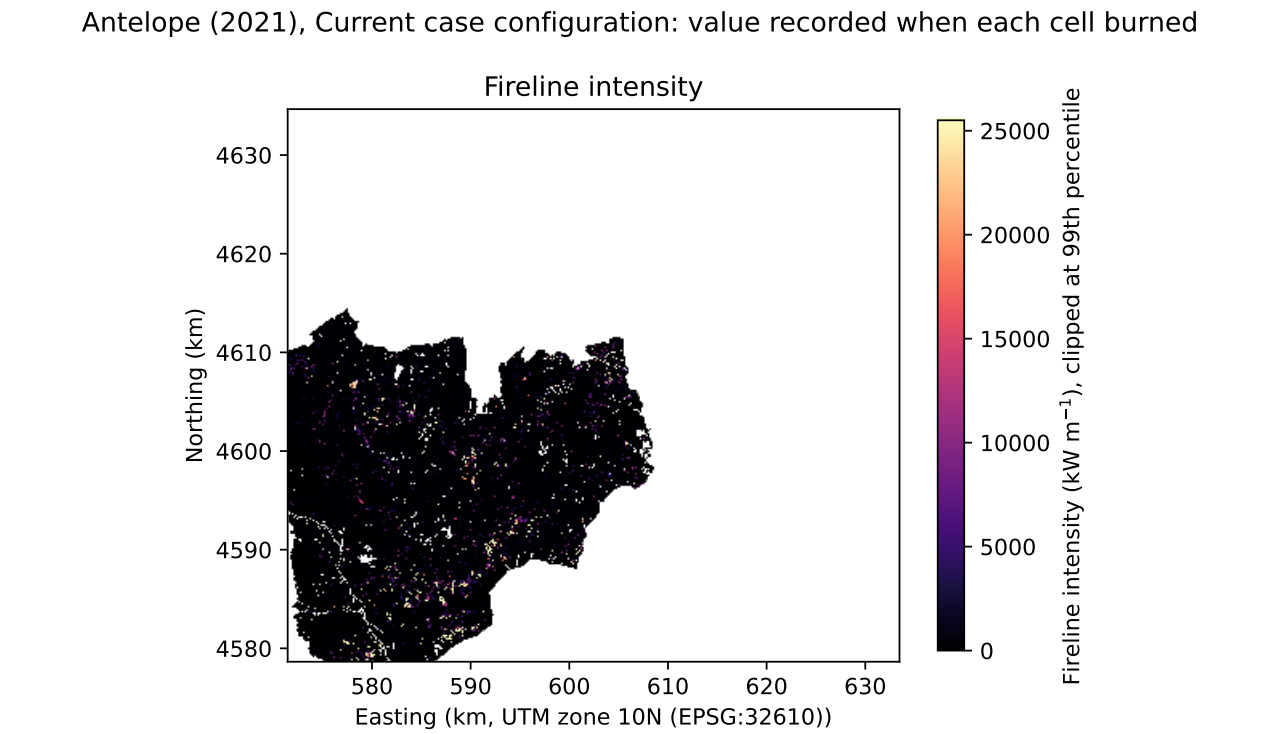


Figure 9: Current case configuration: simulated spread rate and fireline intensity recorded when each cell burned, for the single simulation member at its final output time; color scales are clipped at the 99th percentile.

8 Integrated statistics and calculated metrics

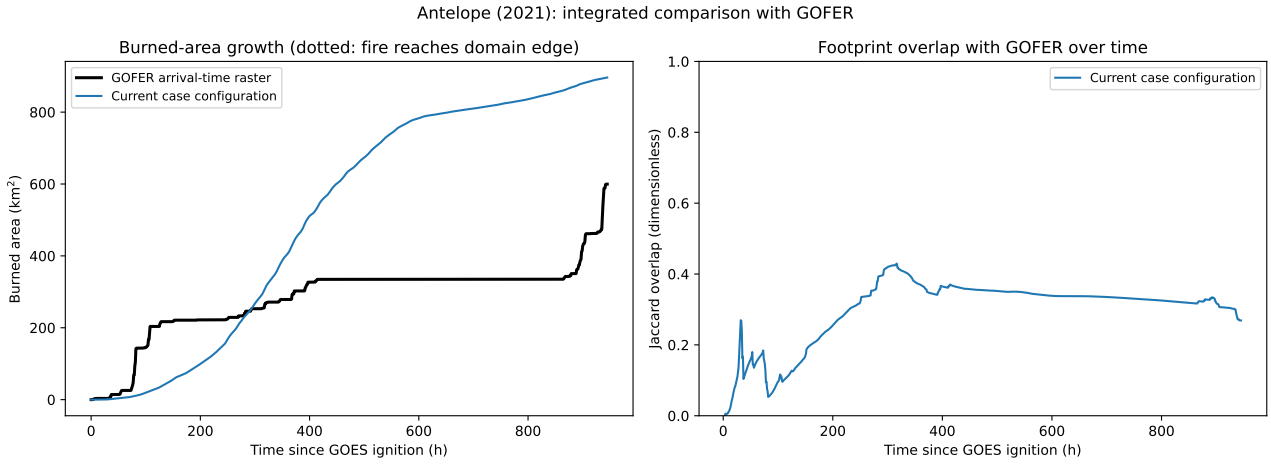


Figure 10: Observed (GOFER) and simulated burned-area growth, and the Jaccard overlap of each simulation with the GOFER footprint, evaluated hourly since the GOES ignition hour. Dotted lines mark the time each simulated fire first reaches the model-domain boundary.

Current case configuration

Time (h)	GOFER (km^2)	Simulated (km^2)	Area ratio	Jaccard	Precision	Recall	F_1
24	3.4	0.329	0.0965	0.097	1	0.097	0.18
48	14.2	3.01	0.212	0.15	0.75	0.16	0.26
72	25.6	7.57	0.296	0.18	0.66	0.2	0.3
945	600	896	1.49	0.27	0.35	0.53	0.42

Table 3: Footprint metrics of the current case configuration against the GOFER-derived footprint at fixed times since the GOES ignition hour, evaluated on the 30-m model grid over valid landscape cells. Time is the comparison instant; areas are burned areas through that instant.

Quantity	Value
Cells burned in both by 945 h	352,276
Mean arrival-time difference, simulated minus GOFER (h)	64.3
Median arrival-time difference (h)	107
Mean absolute arrival-time difference (h)	158
Root-mean-square arrival-time difference (h)	177
Simulated end since GOES ignition (h)	945
Propagation stalled before the planned end	no
First contact with domain boundary (h)	none
Final simulated area (ac)	221,476
Final crown-fire area (ac)	25,416

Table 4: Arrival-time and integrated statistics of the current case configuration. Arrival-time differences use cells burned in both the simulation and the GOFER observation by the final comparison time.

Recall exceeds precision at the final comparison time, so most of the observed footprint is reached but a larger share of the simulated footprint lies outside it: the simulation overgrows the observation. The simulated burned area is 1.5 times the observed area at 945 h. Where both burned, the simulated fire arrives on average 64.3 h later than the observed fire (mean absolute difference 157.8 h).

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Antelope reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **CHARACTERIZED**, with 1 simulation result(s) evaluated. A defensible acceptance decision

requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.